

MEMORIAL ONCOLOGY & PATHOLOGY
INSTITUTE

DEPARTMENT OF MOLECULAR DIAGNOSTICS & CLINICAL
HISTOPATHOLOGY

CAP ACCREDITED #7192834

CLIA ID: 36D0891234

NCCN GUIDELINE ALIGNED

SURGICAL PATHOLOGY & COMPREHENSIVE BIOMARKER REPORT

FINAL VERIFIED REPORT

Patient Name:	Eleanor Vance	Medical Record (MRN):	ONCO-2026-88421
Age / Sex:	52 Years / Female	Date of Birth:	14-Mar-1974
Menopausal Status:	Post-menopausal	Accession Number:	SP-2026-09412-B
Referring Oncologist:	Dr. Rajesh Mehta, MD	Date of Collection:	14-Aug-2026
Specimen Source:	Left Breast Core Biopsy & SLNB	Report Verified Date:	18-Aug-2026

FINAL PATHOLOGICAL DIAGNOSIS

- **Primary Tumor:** Invasive Ductal Carcinoma (Infiltrating Ductal Carcinoma, NOS), Left Breast (Upper Outer Quadrant).
- **Tumour Size:** 2.8 cm in greatest unidimensional diameter.
- **Histologic Grade:** Nottingham Grade 2 (Moderately Differentiated) — Tubule Score 3, Nuclear Pleomorphism 2, Mitotic Count 2 (Total Score: 7/9).
- **Pathological Stage:** Stage IIA (pT2 N0 M0, AJCC 8th Edition).
- **Lymph Node Status:** 0 / 3 Sentinel Lymph Nodes Positive (No regional nodal metastasis detected; 0 involved nodes).
- **Lymphovascular Invasion (LVI):** Not identified. Surgical margins: Negative (> 5mm clear).

IMMUNOHISTOCHEMISTRY (IHC) & CORE BIOMARKERS

BIOMARKER / ASSAY	STATUS / RESULT	QUANTIFICATION / DETAILS	CLINICAL INTERPRETATION
Estrogen Receptor (ER)	Positive	90% tumor cell nuclei (Allred Score 8/8)	Strong Hormone Sensitivity
Progesterone Receptor (PR)	Positive	75% tumor cell nuclei (Allred Score 7/8)	Positive (Endocrine responsive)
HER2 / neu Protein (IHC)	Negative	Score 1+ (Incomplete membrane staining)	HER2 Negative (ASCO/CAP)
HER2 Dual-ISH / FISH	Negative	Ratio 1.18, Mean copy 2.3 signals/cell	Not Amplified (Confirms HER2-)
Ki-67 Proliferation Index	14%	MIB-1 monoclonal antibody index	Low Proliferation (< 20% threshold)
Tumour Infiltrating Lymphocytes (TILs)	15%	Stromal compartment evaluation	Moderate immune infiltration
PD-L1 Expression (22C3)	Negative	CPS < 1 (Combined Positive Score)	No PD-L1 overexpression

MOLECULAR GENETICS & MULTIGENE PROFILING

GENOMIC TEST	RESULT / SCORE	RISK CATEGORY	THERAPEUTIC IMPLICATION
Oncotype DX Recurrence Score	Score: 16	Low Risk (RS 0–25)	No significant chemotherapy benefit; endocrine therapy indicated.
MammaPrint 70-Gene Profile	Low Risk	Genomic Low Risk	Favorable prognosis; supports de-escalation of adjuvant chemotherapy.
Germline BRCA1 Mutation	Negative	Wild-Type (Not Detected)	No pathogenic germline BRCA1 variant detected.
Germline BRCA2 Mutation	Negative	Wild-Type (Not Detected)	No pathogenic germline BRCA2 variant detected.

PATIENT HEALTH, CARDIAC & SAFETY BASELINE

PARAMETER	MEASUREMENT	SAFETY EVALUATION
Left Ventricular Ejection Fraction (LVEF)	62%	Normal cardiac function (baseline pre-treatment echocardiogram).
ECOG Performance Status	Score: 0	Fully active, able to perform all daily activities without restriction.
Comorbidities Recorded	Essential Hypertension (Mild, well-controlled)	No renal, hepatic, or severe cardiovascular contraindications.
Current Medications	Amlodipine 5 mg OD, Multivitamin oral daily	No major antineoplastic drug interactions detected.

CLINICAL SUBTYPE INTEGRATION (ST. GALLEN / NCCN)

Molecular Subtype: Luminal A-like (ER+ / PR+ / HER2- / Ki-67 14%)
Algorithmic Confidence: 94% confidence based on concordant ER/PR positivity, validated HER2 negativity, and Ki-67 < 20%.
Primary Recommendation: Adjuvant Endocrine Therapy (Aromatase Inhibitor e.g., Letrozole or Anastrozole for 5–10 years). Adjuvant chemotherapy may be safely omitted based on Oncotype DX score of 16 (TAILORx trial criteria).

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